

Chapter 18: Linear Independence Worksheet

1. Are each of the following statements true or false?

- (a) If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent, so is $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3\}$.
- (b) If $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent, so is $\{\mathbf{v}_1, \mathbf{v}_2\}$.
- (c) Two vectors are linearly dependent if and only one is a scalar multiple of the other.
- (d) The set $\{\mathbf{v}\}$ for a nonzero vector $\mathbf{v}$ in a vector space V is linearly independent.
- (e) If S contains $\mathbf{0}$, then S must be linearly dependent.
- (f) If $\text{span}S$ contains $\mathbf{0}$, then S must be linearly dependent.

Solution.

- (a) True. Consider $a\mathbf{v}_1 + b\mathbf{v}_2 + c(\mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3) = \mathbf{0}$. I can recollect this as $(a + c)\mathbf{v}_1 + (b + c)\mathbf{v}_2 + c\mathbf{v}_3 = \mathbf{0}$. Since $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a linearly independent set, the coefficients must all be zero: $a + c = 0, b + c = 0, c = 0$. $c = 0$ will force $a = 0$ and $b = 0$, so the original set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_1 + \mathbf{v}_2 + \mathbf{v}_3\}$ is also linearly independent.
- (b) False. Consider: $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$. This set is linearly dependent (check!) but $\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ is linearly independent.
- (c) True. If $\mathbf{u}, \mathbf{v}$ are linearly dependent, then there are a, b with $a\mathbf{u} + b\mathbf{v} = \mathbf{0}$ with $(a, b) \neq (0, 0)$. If $a \neq 0$, then $\mathbf{u} = -\frac{b}{a}\mathbf{v}$ (If $a = 0, b \neq 0$, so $\mathbf{v} = -\frac{a}{b}\mathbf{u} = \mathbf{u}$), so one vector is a scalar multiple of the other. If $\mathbf{u} = c\mathbf{v}$, then $\mathbf{u} - c\mathbf{v} = \mathbf{0}$ is a nontrivial linear combination equalling the zero vector, so $\mathbf{u}, \mathbf{v}$ are linearly dependent.
- (d) True. Try to make a linear combination equal to zero: $a_1\mathbf{v} = \mathbf{0}$. The only way this can happen is if $\mathbf{v} = \mathbf{0}$ (we stated it's not) or if $a_1 = 0$.
- (e) True. Any set that contains the zero vector is automatically linearly dependent.
- (f) False. Notice the use of $\text{span}S$ versus S . The span of any set with automatically contain the zero vector (since it takes all linear combinations of the vectors in S , including the trivial), but that doesn't mean that S must contain $\mathbf{0}$ or be linearly independent. For example, $S = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$ is a linearly independent set in $\mathbb{R}^2$, and $\mathbf{0} = 0 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is in the span of S .

2. Given a set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ in $\mathbb{R}^3$, where:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}.$$

- (a) Determine whether the vectors are linearly independent or linearly dependent.
- (b) If they are linearly dependent, write one vector as a linear combination of the others.

Solution.

- (a) To determine whether the vectors are linearly independent or dependent, we can form a matrix A using these vectors as columns:

$$A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}.$$

If we compute the $\det(A) = 0$, we know they are linearly dependent. But for part 2, we want to row reduce it to echelon form. The RREF is in fact:

$$\text{RREF}(A) = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

This tells us: If $a\mathbf{v}_1 + b\mathbf{v}_2 + c\mathbf{v}_3 = \mathbf{0}$, then c is a free variable, $a = c$, and $b = -2c$. In particular, if we set $c = 1$, we have $a = 1$, $b = -2$, so $1\mathbf{v}_1 - 2\mathbf{v}_2 + 1\mathbf{v}_3 = \mathbf{0}$.

- (b) Using our answer for part 1, we can solve $\mathbf{v}_3 = 2\mathbf{v}_2 - \mathbf{v}_1$. Let's verify:

$$\begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} = 2 \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

3. Determine if each of the following sets is linearly independent.

- (a) $\left\{ \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \right\}$ (in $M_{2 \times 2}$)
- (b) $\{x + 1, x^2 - 1, x^2 + x - 3, 4 - x\}$ (in P_2)
- (c) $\left\{ \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 5 \end{bmatrix}, \begin{bmatrix} 3 & 6 & 9 \\ -3 & 1 & 15 \end{bmatrix} \right\}$ (in $M_{2 \times 3}$)
- (d) $\left\{ \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\}$ (in $\mathbb{R}^4$).

Solution.

(a) We set up a linear combination of these matrices and set it equal to the zero matrix:

$$a_1 \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} + a_2 \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} + a_3 \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

This is equivalent to a homogeneous system:

$a_1 + a_2 = 0, 2a_1 + a_3 = 0, a_2 + a_3 = 0, 2a_2 + a_3 = 0$ which can be translated to a matrix and reduced:

$$\begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 2 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 2 & 1 \end{bmatrix}$$

(note: we omit the constants column, since each entry is 0)

This tells us that the only solution is $a_1 = 0, a_2 = 0, a_3 = 0$, so this set is linearly independent.

(b) We set up a linear combination of these vectors and set it equal to the zero polynomial:

$$a_1(x + 1) + a_2(x^2 - 1) + a_3(x^2 + x - 3) + a_4(4 - x) = 0$$

Collecting like terms, we see that the system

$a_1 + a_3 - a_4 = 0, a_1 - a_2 - 3a_3 + 4a_4 = 0, a_2 + a_3 = 0$ has four unknowns and 3 equations. For a homogeneous system, this guarantees a nontrivial solution (in fact, infinitely many nontrivial solutions), so this set of vectors is not linearly independent. In other words, it's linearly dependent.

Note: We could also have applied that fact that P_2 can be spanned by 3 vectors (ex: $\{1, x, x^2\}$), so a set of 4 vectors is automatically linearly dependent.

(c) Since there are only two vectors, we can check if they are scalar multiples of each other.

We see that $3 \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 3 & 6 & 9 \\ -3 & 0 & 15 \end{bmatrix}$ almost works, but the 2, 2-entry of the second matrix is 1, not 0. Thus, these matrices are linearly independent.

(d) we set up a linear combination of these vectors and set it equal to the zero vector:

$$a_1 \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix} + a_2 \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \end{bmatrix} + a_3 \begin{bmatrix} 0 \\ 0 \\ 2 \\ 0 \end{bmatrix} + a_4 \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$